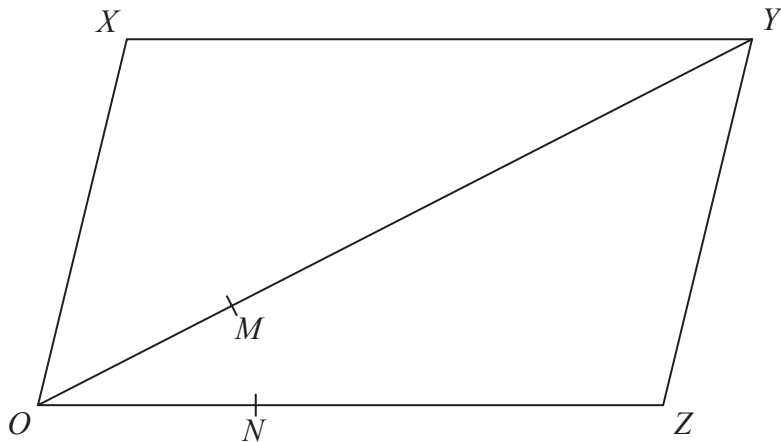


15 $OXYZ$ is a parallelogram.



$$\vec{OY} = \mathbf{a} \text{ and } \vec{OZ} = \mathbf{b}$$

M is the point on OY such that $OM:MY = 1:3$

N is the point on OZ such that $ON:NZ = 1:2$

Work out the ratio $XN:MN$

You must show all your working.

(Total for Question 15 is 4 marks)